

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\hbar\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1^\dagger\right)^2}{\epsilon} + C_{S_1^\dagger S_1}S_1S_1^\dagger + D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$ $-C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger$ $-\frac{2C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1^\dagger}{\epsilon} + \frac{1}{2}S_1^2\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y}_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$ $-\hbar\overline{y}_{1,i_1,r_1}S_1^2y_{1,i_1,r_1}\left(S_1^\dagger\right)^2\lambda_{S_1}^{(1)}$		$+\frac{2\hbar\overline{y}_{2,i_1,r_1}y_{2,i_2,r_2}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}^\dagger\right)^2}{3\epsilon} + C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}}$ $-\frac{2C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y}_{2,i_1,r_1}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{3\epsilon} + \frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^1\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^2\lambda_{S_2}^{(2)}$ $-\frac{2\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_3},j_1,j_4,j_2}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
cont.										
cont.										
cont.										
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$ $-\frac{1}{2}C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$ $-\frac{C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}S_{1,j_1}\overline{S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}}{\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}$ $-\frac{\hbar\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-\frac{1}{2}C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $-\frac{C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y}_{2,i_1,r_1}y_{2,i_2,r_2}S_{2,j_1}\overline{S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}$ $-\frac{\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}$
cont.										
cont.										
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}^\dagger$ $-\frac{1}{2}\sqrt{3}C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger$ $-\frac{\sqrt{3}C_{S_1^\dagger S_1}\overline{\hbar y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}S_{1,j_1}\overline{S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger}}{3\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}$ $-\frac{\sqrt{3}\hbar\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y}_{1,i_1,r_1}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_{1,j_1},j_3,j_2,j_4}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}^\dagger$ $-\frac{1}{2}\sqrt{3}C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $-\frac{\sqrt{3}C_{S_2^\dagger S_2}\overline{\hbar y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y}_{2,i_1,r_1}y_{2,i_2,r_2}S_{2,j_1}\overline{S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}$ $-\frac{\sqrt{3}\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y}_{2,i_1,r_1}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_{2,j_1},j_3,j_2,j_4}^1\lambda_{S_2}^{(1)}$
cont.										
cont.										

[illegible]

Field combinations 2 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$		$-\frac{\hbar B_{\mu_1\mu_2}^2}{2\epsilon}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{2\epsilon}$

Field combinations 3 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$S_1 H$	$S_1 S_2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_2} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1^\dagger^\dagger S_{2,j_1}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger}{\sqrt{3}}$	$+\frac{4\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_2,r_2} y_{2,i_1,r_2} S_1^\dagger^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$
								$+\frac{2\hbar\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$+\frac{4\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1} S_1 S_{2,j_1} y_{1,i_1,r_3} y_{2,i_2,r_2} S_1^\dagger^\dagger S_{2,j_1}^\dagger}{\sqrt{3}}$
cont.								$+\frac{2\hbar\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+S_1 S_{2,j_2} S_1^\dagger^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}$
cont.									$-\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1^\dagger^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.										$-\frac{2\hbar\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\bar{y}_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1^\dagger^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_2,j_1} {}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1}^* H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$	$+\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger}{\epsilon}$
								$+\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-2\hbar\bar{y}_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y1,a_1,j_2} \mathcal{I}_{y1,a_2,j_1}^* H_{a_1}^\dagger S_{1,j_1}^\dagger$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger}{\epsilon}$
								$+\frac{2\hbar\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^1\lambda_{H1}^{(1)}$	$-\frac{2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+\frac{2\hbar\lambda_{T3} L_{a_3,i_1} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^1\lambda_{H1}^{(1)}}{\epsilon}$	$+2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_2} y_{2,i_1,r_2} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger$
cont.									$-\frac{1}{2}\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^1\lambda_{H1}^{(1)}$	$+2\hbar\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger$
cont.									$+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^2\lambda_{H1}^{(2)}$	$+S_{1,j_4} S_{2,j_3} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.									$-\frac{\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^2\lambda_{H1}^{(2)}}{\epsilon}$	$-\frac{\hbar\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.									$-\frac{1}{2}\hbar\bar{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2} {}^2\lambda_{H1}^{(2)}$	$-\frac{1}{2}\hbar\bar{y}_{1,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4} {}^1\lambda_{12}^{(1)}$
cont.										$-\frac{1}{2}\hbar\bar{y}_{2,i_1,r_1} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_2$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	H	S_1	S_2
													$+ \frac{\overline{h y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_1,j_3,j_4,j_2}^2 \lambda_{12}^{(2)}}{\sqrt{3}}$
cont.													$- \frac{2 \overline{h y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_1,j_3,j_4,j_2}^2 \lambda_{12}^{(2)}}{\sqrt{3} \epsilon}$
													$- \frac{\overline{h y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_1,j_3,j_4,j_2}^2 \lambda_{12}^{(2)}}{\sqrt{3}}$
													$+ S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_3,j_1,j_4,j_2}^3 \lambda_{12}^{(3)}$
cont.													$- \frac{2 \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_3,j_1,j_4,j_2}^3 \lambda_{12}^{(3)}}{\sqrt{3} \epsilon}$
													$- \frac{\overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_3,j_1,j_4,j_2}^3 \lambda_{12}^{(3)}}{\sqrt{3}}$
													$- \frac{2 \overline{h y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_3,j_1,j_4,j_2}^3 \lambda_{12}^{(3)}}{\sqrt{3} \epsilon}$
cont.													$- \frac{\overline{h y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger T_{12,j_3,j_1,j_4,j_2}^3 \lambda_{12}^{(3)}}{\sqrt{3}}$

Field combinations 4 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_2 H$	$S_2 F$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & -2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_1^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_1^\dagger \right)^2 \\ & -2C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_1 y_{1,i_1,r_1} S_1^\dagger} \\ & -2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_1^2 y_{1,i_1,r_1} \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)} \end{aligned}$	$\begin{aligned} & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3\epsilon}} \\ & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & +\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}{\epsilon} \\ & +2i\sqrt{\frac{2}{3}}h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^1 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)} \\ & -\frac{2h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3\epsilon}} \\ & -\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & +\frac{2}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2 \\ & -\frac{2C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}}{\sqrt{3}} \\ & -\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_3,j_1,j_4,j_2}^1 \lambda_{S_2}^{(1)}}}{\sqrt{3}} \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_3,j_1,j_4,j_2}^2 \lambda_{S_2}^{(2)}}}{\sqrt{3}} \end{aligned}$
cont.										
cont.										
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\begin{aligned} & -C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger} \\ & -h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}} \end{aligned}$	$\begin{aligned} & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y2,a_2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon} \\ & -2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} T_{y2,a_2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)} \\ & -\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}}{\epsilon} \\ & -\frac{1}{2}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)} \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)} \\ & -\frac{h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}}{\epsilon} \\ & -\frac{1}{2}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)} \end{aligned}$	$\begin{aligned} & -C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}} \end{aligned}$
cont.										
cont.										
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger} \\ & -\frac{5}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}} \end{aligned}$	$\begin{aligned} & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3\epsilon}} \\ & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3\epsilon}} \\ & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1}{\sqrt{3}} \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)} \\ & -\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)} \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)} \\ & -\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}}{\epsilon} \\ & -\frac{1}{2}\sqrt{3}h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)} \end{aligned}$	$\begin{aligned} & -\sqrt{3}C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -\frac{5}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}} \end{aligned}$
cont.										
cont.										
cont.										
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\begin{aligned} & +\frac{2}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger \right)^2 \\ & -\frac{2C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}} \\ & -\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_3,j_1,j_4,j_2}^1 \lambda_{S_1}^{(1)}}}{\sqrt{3}} \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_3,j_1,j_4,j_2}^2 \lambda_{S_1}^{(2)}}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon} \\ & -2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2}^1 \lambda_{H2}^{(1)} \\ & -\frac{2h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2}^1 \lambda_{H2}^{(1)}}{\epsilon} \\ & -h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2}^1 \lambda_{H2}^{(1)} \end{aligned}$	$\begin{aligned} & -2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2 \\ & -2C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_2} y_{2,i_1,r_1} S_{2,j_1}^\dagger} \\ & -2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_2}^2 y_{2,i_1,r_1} \left(S_{2,j_1}^\dagger \right)^2 \lambda_{S_2}^{(1)}} \end{aligned}$
cont.										
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & +\frac{2}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_2}^2 y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}^\dagger \right)^2 \\ & -\frac{2C_{S_1^\dagger S_1} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}}{\sqrt{3}} \\ & -\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_3,j_1,j_4,j_2}^1 \lambda_{S_1}^{(1)}}}{\sqrt{3}} \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_3,j_1,j_4,j_2}^2 \lambda_{S_1}^{(2)}}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3\epsilon}} \\ & -\frac{2h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & +\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}{\epsilon} \\ & +2i\sqrt{\frac{2}{3}}h\overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^1 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)} \\ & -\frac{2h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3\epsilon}} \\ & -\frac{2h\overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & +\frac{2}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_2^2 y_{2,i_1,r_2} y_{2,i_2,r_1} \left(S_{2,j_1}^\dagger \right)^2 \\ & -\frac{2C_{S_1^\dagger S_2} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}}{\sqrt{3}} \\ & -\frac{4}{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_3,j_1,j_4,j_2}^1 \lambda_{S_2}^{(1)}}}{\sqrt{3}} \\ & -\frac{2h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_3,j_1,j_4,j_2}^2 \lambda_{S_2}^{(2)}}}{\sqrt{3}} \end{aligned}$
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	F	S_2	H	S_2	F
cont.											$-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}{}^2 \lambda_{H2}^{(2)}}{\sqrt{3}}$		

Field combinations 5 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F W_{\mu\nu}$	$H \ell e$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$		$+\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{1,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{2,i_1,r_1}Y_{e,i_3,i_2}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$
cont.										
cont.										
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\frac{\sqrt{3}\hbar y_{1,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{2,i_3,r_1}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F B_{\mu\nu}$	$F W_{\mu\nu}$	$H \ell e$
cont.										$+\frac{3}{8}\overline{h y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{h y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$
										$+\frac{3}{8}\sqrt{3}\overline{h y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\overline{h y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$
cont.										$+\frac{3}{8}\overline{h y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\sqrt{3}\overline{h y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$+\frac{3}{8}\sqrt{3}\overline{h y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\overline{h y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{h y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$
cont.										$+\frac{\sqrt{3}\overline{h y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{h y_{1,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$
cont.										$+\frac{\sqrt{3}\overline{h y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\overline{h y_{2,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$

Field combinations 6 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 H$	$S_1 S_2 F$	$S_1 S_2 \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda T_3 S_{2,j_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}$ $-\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{\lambda T_3 S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\epsilon}$ $-\frac{1}{2} \hbar \overline{y_{1,i_1,r_1}} \overline{\lambda T_3 S_{2,j_1} y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right)}{\sqrt{3}} \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_1{}^\dagger S_{2,j_1}{}^\dagger}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} S_1 S_{2,j_2} y_{1,i_1,r_1} S_1{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{12,j_2,j_1}{}^\dagger \lambda_{12}^{(1)}}$ $-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} S_1 S_{2,j_2} y_{2,i_1,r_1} S_1{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{12,j_2,j_1}{}^\dagger \lambda_{12}^{(1)}}$	$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}}{M_F}$ $-\frac{\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}}{4\epsilon M_F}$ $-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}}{8M_F}$
								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$	$-\frac{\hbar y_{1,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $-\frac{1}{2} \hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger$ $-\frac{\hbar y_{2,i_1,r_1} H_{a_1} H_{a_2} S_1 y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{2\sqrt{3}}$ $+\frac{H_{a_1} H_{a_2} S_1 \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}{\hbar \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_1 y_{1,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} \lambda T_3 S_{2,j_1}{}^\dagger}$		$-\frac{3\hbar y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I} T_3^*_{a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$
cont.								$-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3^*_{a_1,a_2,j_1} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda T_3 S_{2,j_1} y_{2,i_1,r_1} \mathcal{I} T_3$		

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	S_1	S_2	ℓ
cont.																$+ \frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger$
																$+ \frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger$
																$+ \frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I} T \tau_{a_1,a_2,j_1,j_2}^* S_{2,j_1}^\dagger$
cont.																$- \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_3} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$- \frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_3} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{9\sqrt{3}h\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.																$+ \frac{3\sqrt{3}h\overline{L_{a_1,i_1}}}{8M_F} C Pr \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$- \frac{\sqrt{3}L_{a_1,i_1}}{2M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$- \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.																$- \frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.																$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{4\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.																$+ \frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
																$+ \frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{8\epsilon M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger$
cont.																$+ \frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}}{16M_F} C Pr \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	S_1	S_2	ℓ
cont.																$\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}CPrL_{a_2,i_2}^{\dagger}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_2^{\dagger}}{8M_F}$ $+\frac{\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{M_F}$
																$\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_1}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$ $+3\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_1}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$
cont.																$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{2\sqrt{3}M_F}$
																$\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$ $+3\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{1,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$
cont.																$\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$
																$\frac{3h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$ $-\frac{h\overline{y}_{1,i_3,r_1}\overline{y}_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$
cont.																$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$
																$\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{8M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{4\epsilon M_F}$
cont.																$\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2y_{2,i_3,r_2}IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}}{2M_F}$ $-\frac{h\overline{L}_{a_1,i_1}CPrL_{a_2,i_2}^{\dagger}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_2^{\dagger}\lambda_{12}^{(1)}}{\epsilon M_F}$
																$\frac{h\overline{L}_{a_1,i_1}CPrL_{a_2,i_2}^{\dagger}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}IT_{3,a_1,a_2,j_1}S_2^{\dagger}\lambda_{12}^{(1)}}{M_F}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}\lambda_{12}^{(1)}}{\epsilon M_F}$
cont.																$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}\lambda_{12}^{(1)}}{M_F}$ $+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}^TCPrL_{a_1,i_1}S_2IT_{3,a_1,a_2,j_1}S_{1,j_1}^{\dagger}\lambda_{12}^{(1)}}{M_F}$
																$+\frac{\overline{\lambda T_3}S_{2,j_2}IT_{3,a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y}_{1,i_1,r_1}\overline{\lambda T_3}S_{2,j_2}y_{1,i_1,r_1}IT_{3,a_1,a_2,j_1,j_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}S_{1,j_1}^{\dagger}}{2\sqrt{3}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$									$-\frac{4}{3}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	S_1	S_2	ℓ
																$\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{4\epsilon MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
cont.																$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}L_{a_2,i_2}}{8MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{6MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{6MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
cont.																$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{4\epsilon MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
cont.																$-\frac{3h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}L_{a_2,i_2}}{8MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{4\epsilon MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}L_{a_2,i_2}}{8MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
cont.																$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}{3\epsilon MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}L_{a_2,i_2}}{6MF} \frac{T}{C} \frac{P_L}{L_{a_1,i_1}} \frac{S_{2,j_1}}{y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}}^\dagger$
																$-\frac{h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{2\sqrt{2}\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
cont.																$-\frac{3h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
																$-\frac{\sqrt{6}hL_{a_1,i_1}}{\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
																$-\frac{\sqrt{\frac{2}{3}}hL_{a_1,i_1}}{MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
cont.																$+\frac{\sqrt{\frac{2}{3}}L_{a_1,i_1}}{MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
																$-\frac{h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{2\sqrt{2}\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
																$-\frac{3h\overline{y}_{2,i_3,r_1}L_{a_1,i_1}}{4\sqrt{2}\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_2}}{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}y_{2,i_2,r_1}T\tau_{a_1,a_2,j_2,j_1}^*S_{2,j_1}}^\dagger$
cont.																$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1}L_{a_1,i_1}}{3\epsilon MF} \frac{T}{C} \frac{P_R}{L_{a_2,i_2}} \frac{S_{1,j_$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	H	S_1	S_2	F	S_1	S_2	ℓ
																$+ \frac{\sqrt{\frac{2}{3}} h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\epsilon M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}$
cont.																$+ \frac{\sqrt{\frac{2}{3}} h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}$ $- \frac{h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{3} \epsilon M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}$ $- \frac{h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{3} M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}$
cont.																$+ \frac{\sqrt{\frac{2}{3}} h \overline{L}_{a_1,i_1}}{\epsilon M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}$ $+ \frac{\sqrt{\frac{2}{3}} h \overline{L}_{a_1,i_1}}{M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}$ $- \frac{h \overline{L}_{a_1,i_1}}{\sqrt{3} \epsilon M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}$ $- \frac{h \overline{L}_{a_1,i_1}}{\sqrt{3} M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}$
cont.																$- \frac{3\sqrt{\frac{2}{11}} h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\epsilon M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{3\sqrt{\frac{2}{11}} h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_2} \mathcal{I} T 3_{a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{11} \epsilon M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{11} M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}$
cont.																$- \frac{3\sqrt{\frac{2}{11}} h \overline{L}_{a_1,i_1}}{\epsilon M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{3\sqrt{\frac{2}{11}} h \overline{L}_{a_1,i_1}}{M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I} T 3_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{h \overline{L}_{a_1,i_1}}{\sqrt{11} \epsilon M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}$ $- \frac{h \overline{L}_{a_1,i_1}}{\sqrt{11} M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}$
cont.																$- \frac{7 h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{23} \epsilon M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}$ $- \frac{7 h \overline{y}_{1,i_1,r_1} \overline{y}_{2,i_2,r_1} \overline{L}_{a_2,i_2}}{\sqrt{23} M_F} {}^T C P_L \overline{L}_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}$ $- \frac{7 h \overline{L}_{a_1,i_1}}{\sqrt{23} \epsilon M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}$ $- \frac{7 h \overline{L}_{a_1,i_1}}{\sqrt{23} M_F} C P_R \overline{L}_{a_2,i_2} {}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}$

Field combinations 7 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F H$	$S_2 F H$	$F H \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_1^\dagger \\ & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_1 y_{1,i_1,r_1} H_{a_1}^\dagger S_1^\dagger \mathcal{I} H_{1,\alpha_1,a_2}{}^{1\lambda(1)}_{H1} \end{aligned}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & +2i\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_2}{}^{1\lambda(1)}_{H2} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_2-1}{}^{2\lambda(2)}_{H2}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{\alpha_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F} \\ & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T3} L_{a_3,i_1} C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{\alpha_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} M_F} \end{aligned}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\begin{aligned} & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{T} y_{1,a_2,j_1}^* H_{a_1}^\dagger S_{1,j_1}^\dagger \\ & -\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H1} \\ & -\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{2\lambda(2)}_{H1} \end{aligned}$	$\begin{aligned} & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} \mathcal{T} y_{2,a_2,j_1}^* H_{a_1}^\dagger S_{2,j_1}^\dagger \\ & -\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H2} \\ & -\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_1-j_2}{}^{2\lambda(2)}_{H2} \end{aligned}$	$\begin{aligned} & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{\alpha_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F} \\ & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T3} L_{a_3,i_1} C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{\alpha_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} M_F} \end{aligned}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & -\frac{4\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H1}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H1}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{\alpha_1,a_2} \epsilon_{a_3,a_4}}{3 M_F} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T3} L_{a_3,i_1} C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{\alpha_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}{}^\dagger}{3 M_F} \end{aligned}$
cont.								$\begin{aligned} & -\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H1} \\ & -\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_1-j_2}{}^{2\lambda(2)}_{H1} \end{aligned}$	$\begin{aligned} & -\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_1-j_2}{}^{1\lambda(1)}_{H2} \\ & -\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_1-j_2}{}^{2\lambda(2)}_{H2} \end{aligned}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -2i\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_2-j_1}{}^{1\lambda(1)}_{H1} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_2-j_1}{}^{2\lambda(2)}_{H1}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_2^\dagger \\ & -2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_1,a_2}{}^{1\lambda(1)}_{H2} \end{aligned}$	$\begin{aligned} & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{\alpha_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F} \\ & +\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T3} L_{a_3,i_1} C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{\alpha_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} M_F} \end{aligned}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}} \\ & -2i\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_2-j_1}{}^{1\lambda(1)}_{H1} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_2-j_1}{}^{2\lambda(2)}_{H1}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} \overline{Y}_{e,i_2,i_3} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger}{\sqrt{3}} \\ & +2i\sqrt{\frac{2}{3}}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} Y_{e,i_2,i_3} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{1,\alpha_1,a_2,j_2-j_1}{}^{1\lambda(1)}_{H1} \\ & -\frac{2\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{2,i_1,r_1} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I} H_{2,\alpha_1,a_2,j_2-j_1}{}^{2\lambda(2)}_{H2}}{\sqrt{3}} \end{aligned}$	$\begin{aligned} & -\frac{2\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y}_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{\alpha_1,a_2} \epsilon_{a_3,a_4}}{M_F} \\ & -\frac{2\sqrt{3}\hbar \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda}_{T3} L_{a_3,i_1} C P_R \overline{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{\alpha_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}{}^\dagger}{M_F} \end{aligned}$

Field combinations 8 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 F \ell$	$F H \ell e$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_1}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1^\dagger}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_1}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1^\dagger}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,y_{1,i_1,r_1}} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_2,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,y_{2,i_1,r_1}} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,y_{2,i_1,r_1}} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	$+\frac{1}{4} \sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,y_{2,i_1,r_1}} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_2,r_1} y_{2,i_3,r_2} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1^\dagger}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_2},r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_2},r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_3},r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_3},r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
cont.								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
								$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	$-\sqrt{3} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{2,j_1}^\dagger$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{1}{2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$+3 \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} g_{S_2,j_2}^2 \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$
								$-\frac{1}{2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \overline{\lambda_{T3} S_{2,j_2}} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$
								$-\frac{1}{2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$
								$-\frac{1}{2} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} y_{1,i_3,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} y_{2,i_1,r_2} y_{2,i_3,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_2},r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_2},r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$+\frac{1}{4} \hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal{I}_{a_2,i_2}^T S_{1,j_1}^\dagger$	
cont.								$-\hbar \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R L_{a_2,i_2}^T S_{1,y_{1,i_1},r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{a_1,i_1}^T \mathcal$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	H	$S_1 S_2 F \ell$	$F H \ell e$
cont.												$-\frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right)}{\sqrt{3} M_F} \overline{y_{1,i_2,r_1} y_{2,i_1,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{x}_{y_{1,i_1},j_1}^* T_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$	
												$-\frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right)}{\sqrt{3} M_F} \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{x}_{y_{1,i_1},j_1}^* T_{y_{2,a_2},j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$	
												$-\frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \overline{x}_{y_{1,i_1},j_1} x_{y_{2,a_2},j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
cont.												$-\frac{h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \overline{x}_{y_{1,i_1},j_1} x_{y_{2,a_2},j_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$					$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{1,i_1,r_1} \overline{x}_{T3 a_{1,a_2},j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$	$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{11} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_3}^\dagger$
												$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} \lambda_{T3} S_{2,j_2} y_{2,i_1,r_1} \overline{x}_{T3 a_{1,a_2},j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$	$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{11} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_3}^\dagger$
												$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} H_{a_1} S_{1,j_2} y_{1,i_1,r_1} \overline{x}_{T3 a_{1,a_2},j_2} j_1 \lambda_{T3} S_{2,j_1}^\dagger}$	$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_3}^\dagger e_{i_2} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}$
cont.												$-\frac{1}{2} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} H_{a_1} S_{1,j_2} y_{2,i_1,r_1} \overline{x}_{T3 a_{1,a_2},j_2} j_1 \lambda_{T3} S_{2,j_1}^\dagger}$	$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_3}^\dagger e_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}$
cont.												$+\frac{9 \sqrt{3} h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_{S_2,i_2}^2 \overline{T_{F3,a_1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.												$+\frac{3 h \log \left(\frac{\hat{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \overline{x}_{T3 a_{1,a_2},j_2} S_{1,j_1}^\dagger}{8 M_F}$	
cont.													

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	H	S_1	S_2	F	ℓ	F	H	ℓ	e								
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \overline{I_{T3,a_1,a_2,j_1}} \lambda_{T3} S_2^\dagger}{\sqrt{3}}$												$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_2,r_1} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
								$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \overline{I_{T3,a_1,a_2,j_1}} \lambda_{T3} S_2^\dagger$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_2} y_{2,i_2,r_1} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$			
								$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \lambda_{T3}^* S_2 y_{1,i_1,r_1} \overline{I_{T3}^*} S_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_2} y_{2,i_1,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{\sqrt{3} M_F}$				$+\frac{1}{4} \sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
cont.								$-h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} \lambda_{T3}^* S_2 y_{2,i_1,r_1} \overline{I_{T3}^*} S_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$												$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{\sqrt{3} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
cont.																											
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										
	$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$												$-\frac{h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}} S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \overline{I_{T3}^*} S_{a_1,a_2,j_1} S_2^\dagger}{4M_F}$				$+\frac{1}{4} h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} Y_{e,i_3,i_2} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	H	$S_1 S_2 F \ell$	$F H \ell e$
												$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_2,r_1} Y_{1,i_3,r_2} Y_{2,i_1,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$	
cont.												$- \frac{2\sqrt{3}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_{1,j_1}^2 Y_{1,i_1,r_1} Y_{1,i_3,r_2} Y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{M_F}$	
												$+ \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_2,r_2} Y_{2,i_1,r_2} Y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_2} Y_{1,i_3,r_2} Y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$	
cont.												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_2} Y_{2,i_2,r_2} Y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{3M_F}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_2} Y_{1,i_2,r_1} Y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_2} Y_{2,i_2,r_1} Y_{2,i_3,r_3} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger}{4M_F}$	
cont.												$+ \frac{\sqrt{\frac{2}{3}}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{X_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
												$+ \frac{\sqrt{\frac{8}{3}}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} Y_{1,i_1,r_1} Y_{2,i_2,r_1} \overline{X_{T3,a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_1} Y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
												$- 3\sqrt{\frac{4}{3}}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \overline{X_{T3,a_1,a_2,j_1,j_2}} S_{1,j_1}^\dagger \lambda_{12}^{(2)}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{M_F}$	
cont.												$- \frac{3\sqrt{\frac{2}{3}}h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} Y_{1,i_1,r_1} Y_{2,i_2,r_1} \overline{X_{T3,a_1,a_2,j_2,j_1}} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
												$- \frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_1} Y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
												$- \frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{Y_{1,i_1,r_1} Y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
cont.												$- \frac{7h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} Y_{1,i_1,r_1} Y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	